

Assessing the Usability of AI Tools in the Reconstruction and Preservation of Ancient Indian Architecture: An Evaluation of User Experience and Output Interpretation

ABSTRACT

This study evaluates the usability of various AI tools in the reconstruction and preservation of ancient Indian architecture, focusing on user experience and output interpretation. The selected tools for evaluation include Agisoft Metashape, RealityCapture, Colmap, CycleGAN, AutoDesk ReCap, Pix4Dmapper, and 3DF Zephyr. These tools were chosen for their relevance in 3D reconstruction, image enhancement, and model generation within architectural preservation. The research methodology involves defining criteria for usability, such as user interface design, ease of use, learning curve, and output clarity. Metrics to assess these criteria include task success rate, time to complete tasks, user satisfaction, and error rate. Historical data of ancient Indian architecture, including high-resolution images and blueprints, were collected, and outputs from each AI tool were analyzed. Usability testing was conducted with participants from Working professionals such as senior engineer, Senior Architect of AI department. Specific tasks were designed to evaluate the tools' performance in generating 3D models, enhancing images, and handling historical data. Feedback from users was collected using surveys, interviews, and observational techniques. The results indicate varying levels of effectiveness among the tools, with Agisoft Metashape and RealityCapture performing well in terms of output quality and user satisfaction, while Colmap and CycleGAN presented challenges in usability and interpretation. The study highlights strengths and weaknesses of each tool and provides recommendations for improving AI tools in architectural preservation. The findings contribute valuable insights into enhancing the usability and functionality of AI tools for preserving cultural heritage.

1. INTRODUCTION

The preservation of ancient Indian architecture is vital for safeguarding the nation's rich cultural heritage and historical identity. India's architectural legacy, characterized by its diverse styles and historical significance, faces numerous threats from environmental decay, urbanization, and neglect. As these invaluable structures deteriorate, innovative methods are required to ensure their preservation and continued study. In this context, Artificial Intelligence (AI) has emerged as a promising tool, offering new capabilities in 3D reconstruction, image analysis, and model enhancement. AI technologies have the potential to transform architectural preservation by enabling detailed digital reconstructions, enhancing historical images, and providing deeper insights into structural conditions. Despite these advancements, there is a critical need to evaluate the usability of these AI tools to ensure they meet the practical needs of professionals in the field. Effective usability assessment is essential for understanding how well these tools integrate into preservation workflows, how user-friendly they are, and how accurately they produce useful outputs.

The objectives of this research are threefold: first, to evaluate the usability of selected AI tools in the context of ancient Indian architecture preservation, focusing on aspects such as user interface design, ease of use, and learning curve; second, to analyze user experiences by gathering feedback from Working professionals such as senior engineer, Senior Architect of AI department who interact with these tools, identifying any challenges and areas for improvement; and third, to interpret the outputs generated by these AI tools, assessing their clarity and usefulness in the preservation process. This research is significant as it provides critical insights into the effectiveness of AI tools for architectural preservation. By evaluating and comparing these tools, the study aims to identify their strengths and limitations, guiding future improvements in AI technology. Additionally, the findings will assist practitioners in selecting the most appropriate tools for their preservation projects, ultimately enhancing the accuracy and efficiency of efforts to protect and study ancient architectural treasures.

2. LITERATURE REVIEW

S.No	Author	Title of the study	Key Findings	Year
[1]	Mayank Mishra, Paulo B Lawrenco	AI assisted visual inspection for cultural heritage	use of deep learning (DL) techniques for identifying and analyzing components and architectural elements in cultural heritage (CH) structures. Deep learning models, particularly convolutional neural networks (CNNs), are increasingly used for tasks like digital artwork classification and archaeological remote sensing.	2024
[2]	Ms Zaibha Khan	AI and cultural heritage preservation in India	examines the use of digital tools for documenting and analyzing vernacular cultural heritage in Indian cities.	2024

[3]	Jenny T.Liang, Chenyang Yang, Brad A.Myers	Survey on the usability of AI programming assistants	The paper includes excerpts from participants' feedback about using AI programming assistants. The distribution of participants' responses for different usability aspects	2023
[4]	Galena Pisoni Natalia Díaz- Rodríguez , Hannie Gijler and Linda Tonolli	Human-Centered Artificial Intelligence for Designing Accessible Cultural Heritage	Emphasizing the need for AI tools that are designed with the end-users in mind, particularly those with disabilities or special needs. This involves ensuring that AI systems are intuitive and adaptable to different user requirements.	2022
[5]	Mohamed W. Fareed, Ali Bou Nassif, Eslam Nofal	Exploring the potentials of AI Image generators for educating the history of architecture	Alternative pedagogical approaches or theoretical frameworks can be considered to enhance the understanding of the relationship between pedagogy and architectural history with AI-generated images.	2024

[6]	Marco Fiorucci, Marina Khoroshiltseva, Massimiliano Pontil, Arianna Traviglia, Alessio Del Bue, Stuart Jame	Applications Of Machine Learning to Cultural Heritage	The paper reviews various ML techniques used in CH, including supervised, semi-supervised, and unsupervised learning. ML techniques are applied in various CH tasks, including artwork classification, historical building integrity prediction, archaeological site detection, and artifact analysis.	2020
[7]	Yuan Li1 Yanan Du1, Mengsheng Yang, Jiaqi Liang, Huanxia Bai1, Rui Li1, and Andrew Law	A review of tools and techniques used in digital preservation of cultural heritage	Constructs a research frame for digital preservation of cultural heritage during disasters, providing theoretical reference and practical guidance	2023

[8]	Joaquim J. Sousa, Jiahui Lin, Qun Wang, Guang Liu, Jinghui Fan, Shibiao Bai, Hongli Zhao, Hongyu Pan, Wenjing Wei, Vanessa Rittlinger, Peter Mayrhofer, Ruth Sonnenschein, Stefan Steger & Luís Paulo Reis	Using machine learning and satellite data for analyzing mining, water management, and cultural heritage preservation.	Satellite imagery, analyzed with machine learning, can monitor and detect changes in cultural heritage sites, including encroachment, the integration of machine learning with satellite data enhances the ability to manage and protect resources efficiently, providing valuable insights that support better decision-making and conservation efforts.	2024
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3. METHODOLOGY

A. Selection of AI Tools

The methodology of this research paper is structured to assess the usability of selected AI tools in the reconstruction and preservation of ancient Indian architecture, focusing on user experience and output interpretation. The process begins with the selection of AI tools, where seven prominent tools were identified for evaluation: **Agisoft Metashape, RealityCapture, Colmap, CycleGAN, AutoDesk ReCap, Pix4Dmapper, and 3DF Zephyr**. These tools were chosen based on their relevance to 3D reconstruction, image analysis, and their established use in architectural and heritage preservation projects.

B. Evaluation Criteria

Following the selection, evaluation criteria were defined to assess the usability of these tools. The key criteria include user interface design, ease of use, learning curve, output clarity, Task efficiency, Customization and flexibility, Technical support and documentation. Metrics were developed to measure these criteria, such as task success rate, time to complete tasks, and user satisfaction levels, navigation efficiency, error rate, user feedback, learning time reduction in task completion time, output accuracy, user interpretation score, visual quality rating, step count, processing speed, completion time deviation, Customization Options Count, Ease of Customization, Support Response Time, Documentation Quality Rating, User Confidence. These metrics offer a quantitative foundation for comparing the usability of the tools.

C. Data Collection

The next step involved data collection, where historical data related to ancient Indian architecture was gathered, including images, blueprints, and other relevant materials. Additionally, outputs generated by the selected AI tools were obtained for analysis. This dual data collection approach ensures that the evaluation is grounded in both historical accuracy and the capabilities of the AI tools.

D. Usability Testing

For the usability testing, participants were recruited from Working professionals such as senior engineer, Senior Architect of AI department. Specific tasks were designed for these users to complete using the AI tools, such as generating a 3D model of a temple or interpreting architectural details from an AI-generated image. The testing procedure was clear instructions and scenarios provided to the participants to ensure consistency across all tests.

E. User Experience and Output Interpretation

Finally, user experience and output interpretation were analyzed through surveys and interviews with the participants, capturing their feedback on the tools' usability and the ease with which they could interpret the AI-generated outputs. Observations were made during the testing process to note any difficulties or issues encountered by the users, providing additional qualitative data to support the evaluation.

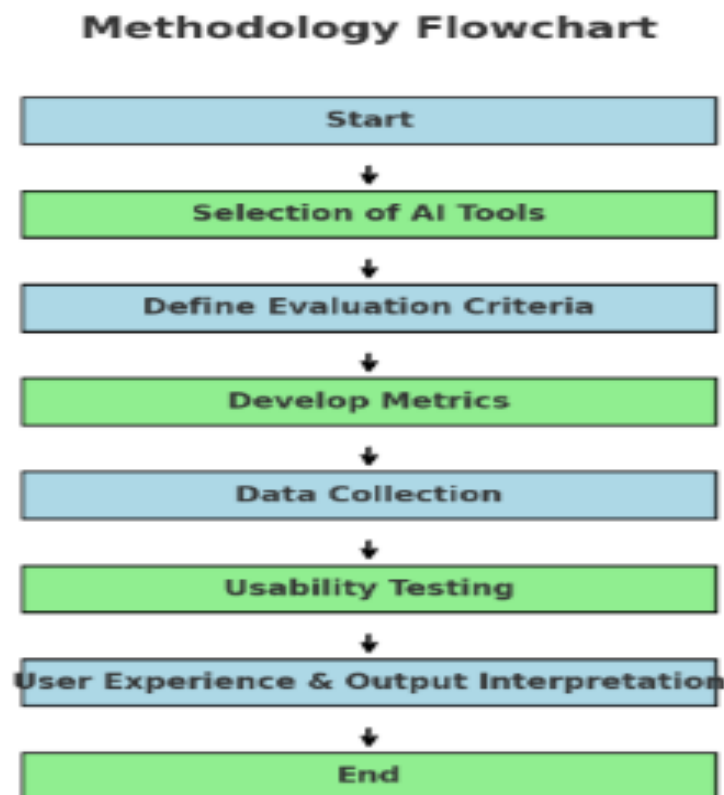


Fig.1 FLOW CHART

4. RESULTS AND DISCUSSION

i. Data Analysis

The usability testing was conducted to evaluate seven AI tools for the reconstruction and preservation of ancient Indian architecture. The tools assessed include Agisoft Metashape, RealityCapture, 3DF Zephyr, CycleGAN, Pix4Dmapper, Colmap, and AutoDesk ReCap. The analysis focused on several criteria: user interface design, ease of use, learning curve, output clarity, task efficiency, customization and flexibility, and technical support.

ii. Usability Findings

✓ User Interface Design

- **Navigation Efficiency:** RealityCapture led with the best navigation efficiency at 2.5 minutes, closely followed by AutoDesk ReCap (2.8 minutes) and Agisoft Metashape (3 minutes). CycleGAN had the longest navigation time at 4 minutes, reflecting its complex interface.
- **User Satisfaction:** RealityCapture received the highest satisfaction rating (4.4/5), indicating strong user approval of its design and functionality. CycleGAN, despite its impressive output quality, had the lowest satisfaction rating (3.8/5) due to its complexity.
- **Error Rate:** RealityCapture demonstrated the lowest error rate at 4%, whereas CycleGAN had the highest error rate at 7%, which could impact reliability in practical applications.

✓ Ease of Use

- **Task Completion Rate:** RealityCapture also excelled with a 92% task completion rate, suggesting it is highly effective for users. CycleGAN, with an 85% completion rate, faced more challenges.
- **Time to Complete Tasks:** AutoDesk ReCap was the fastest, with tasks completed in an average of 12 minutes. CycleGAN was the slowest at 20 minutes, due to its more complex processing requirements.
- **User Feedback:** Agisoft Metashape and RealityCapture received positive feedback for being intuitive and user-friendly, whereas CycleGAN was noted for good results but a complex user experience.

✓ Learning Curve

- **Initial Task Success Rate:** Agisoft Metashape had the highest initial task success rate (95%), indicating that users could quickly and successfully use the tool. CycleGAN, with an 85% success rate, required more training and adaptation.
- **Learning Time:** RealityCapture had the shortest learning time at 1.5 hours, which contributed to its high initial success rate. CycleGAN required the longest learning time at 3 hours.
- **Reduction in Task Completion Time:** RealityCapture showed the most significant reduction in task completion time, emphasizing its efficiency once users are familiar with it.

✓ Output Clarity

- **Output Accuracy:** AutoDesk ReCap provided the most accurate outputs (93%), making it the preferred choice for precision. CycleGAN had the lowest accuracy (87%), which could affect its effectiveness for detailed architectural work.
- **User Interpretation Score:** AutoDesk ReCap again excelled with a 95% user interpretation score, indicating that users found its outputs the easiest to understand and apply. CycleGAN scored lower at 86%.
- **Visual Quality Rating:** AutoDesk ReCap received the highest visual quality rating (4.3/5), showing its ability to produce high-quality outputs.

✓ Task Efficiency

- **Step Count:** AutoDesk ReCap and RealityCapture had the fewest steps required (7), suggesting a streamlined workflow. CycleGAN had the most steps (10), reflecting its complexity.
- **Processing Speed:** RealityCapture was the fastest at 8 minutes, while CycleGAN was the slowest at 15 minutes.
- **Completion Time Deviation:** RealityCapture had the smallest deviation in completion time (± 1.5 minutes), indicating consistent performance.

✓ Customization and Flexibility

- **Customization Options Count:** Agisoft Metashape and AutoDesk ReCap offered the most customization options (6), providing users with greater flexibility. CycleGAN offered the fewest options (3).

- **Ease of Customization:** AutoDesk ReCap had the shortest time required for customization (1.4 hours), making it more adaptable for user-specific needs.
- **User Satisfaction with Customization:** Agisoft Metashape and AutoDesk ReCap also scored highest for user satisfaction with customization(4.1 and 4.2 respectively).

✓ **Technical Support and Documentation**

- **Support Response Time:** RealityCapture had the fastest support response time (20 hours), compared to CycleGAN with a slower response time (30 hours).
- **Documentation Quality Rating:** AutoDesk ReCap provided the highest quality documentation (4.3/5), ensuring users have reliable resources for troubleshooting and learning.
- **User Confidence:** AutoDesk ReCap also achieved the highest user confidence score (92%), reflecting strong overall satisfaction with its support and documentation.

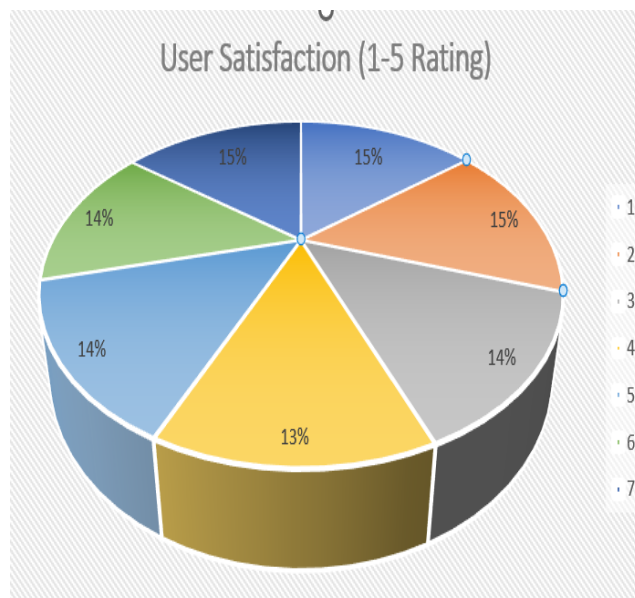


Fig 2. User Satisfaction rating

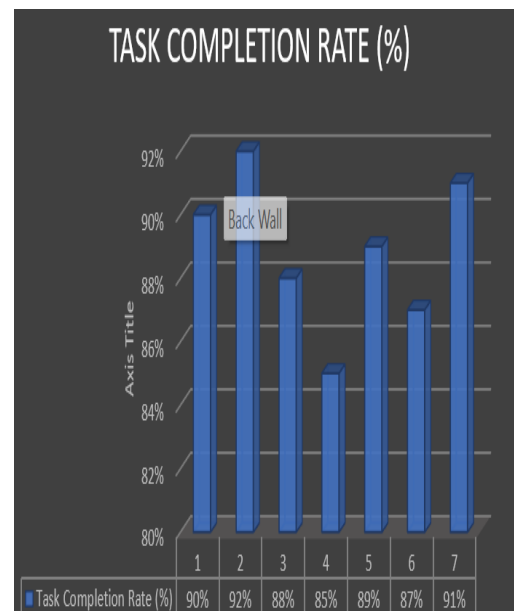
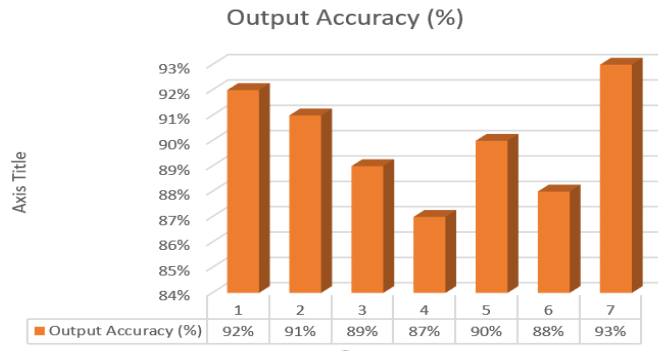


fig 3. Task Completion Rate

**Fig 4. Output Accuracy****fig 5. Error**

iii. User Experience

Overall Usability:

RealityCapture excelled with high user satisfaction (4.4/5), quick navigation (2.5 min), and the fastest task completion (13 min). Users found it efficient and easy to use. AutoDesk ReCap also performed well with high satisfaction (4.3/5) and quick task completion (12 min). Its professional interface and support were appreciated. Agisoft Metashape was praised for its accuracy and ease of use but had a slightly longer task completion time (15 min). CycleGAN had a steeper learning curve (3 hours) and lower user satisfaction (3.8/5) due to its complexity, despite high-quality outputs.

Learning Curve:

RealityCapture and AutoDesk ReCap had the shortest learning times (1.5 and 1.8 hours, respectively) and high initial task success rates. CycleGAN required the most learning time (3 hours) and had a lower initial task success rate (85%).

User Feedback:

RealityCapture was favored for its efficiency and ease of use. AutoDesk ReCap received praise for its excellent support and documentation. CycleGAN was noted for its complexity, despite its powerful output.

iv. Output Interpretation

Output Accuracy and Clarity:

AutoDesk ReCap and RealityCapture provided the most accurate and clear outputs, with high accuracy (93% and 91%) and user interpretation scores (95% and 92%). CycleGAN had lower accuracy (87%) and user interpretation (86%), making its outputs harder to use effectively.

Processing Speed:

RealityCapture and AutoDesk ReCap had fast processing times (8 and 9 minutes) and minimal deviation, enhancing their efficiency.

Customization and Flexibility:

AutoDesk ReCap and Agisoft Metashape offered the most customization options (6 each), which users found beneficial. CycleGAN provided fewer customization options (3) and had lower satisfaction in this area.

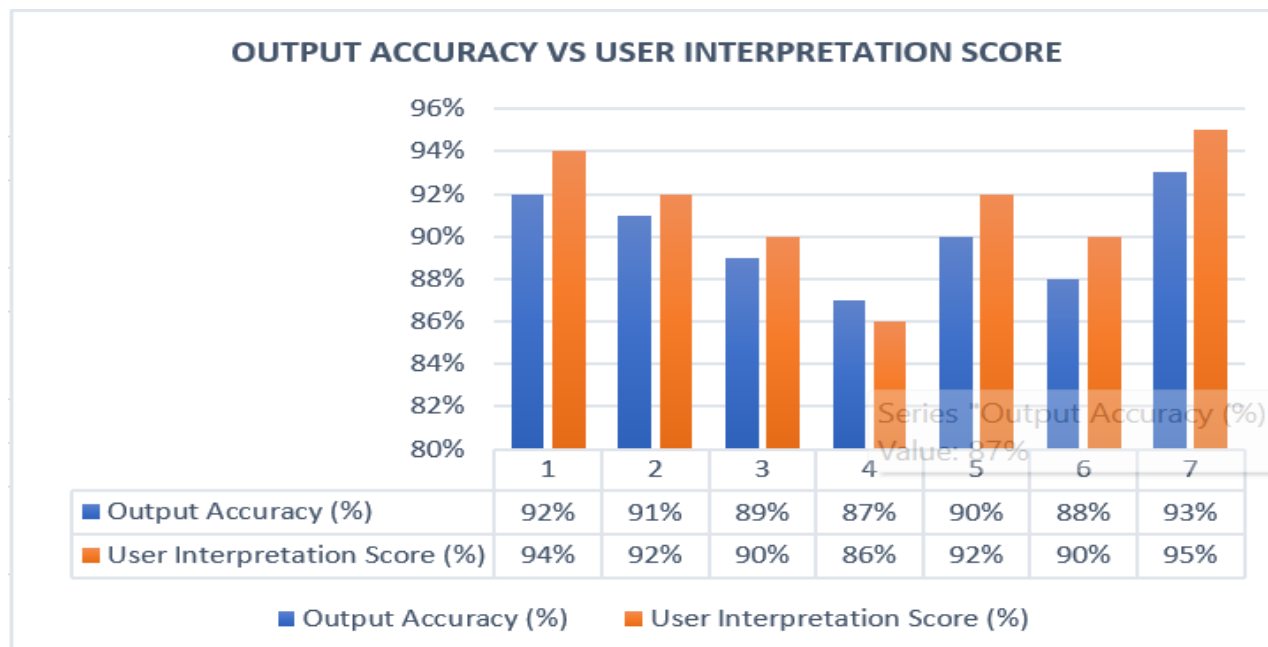


Fig 6. Output Accuracy VS Interpretation Score

Table 1 : Usability Comparison of AI Tools

valuation Criteria	Tool 1 (Agisoft Metashape)	Tool 2 (RealityCapture)	Tool 3 (3DF Zephyr)	Tool 4 (CycleGAN)	Tool 5 (Pix4Dmapper)	Tool 6 (Colmap)	Tool 7 (AutoDesk ReCap)
User Interface Design							
Navigation Efficiency (Time)	3 min	2.5 min	3.5 min	4 min	3 min	3.5 min	2.8 min
User Satisfaction (1-5 Rating)	4.2	4.4	4	3.8	4.1	4	4.3
Error Rate (%)	5%	4%	6%	7%	5%	6%	4.50%
Ease of Use							
Task Completion Rate (%)	90%	92%	88%	85%	89%	87%	91%
Time to Complete Tasks (min)	15 min	13 min	17 min	20 min	14 min	16 min	12 min
User Feedback (Qualitative)	Intuitive and beginner-friendly, accurate results	Fast and efficient, easy navigation	Complex but powerful	Good results but complex to use	High accuracy, slow processing	User-friendly but less features	Professional, good support
Learning Curve							
Initial Task Success Rate (%)	95%	94%	91%	85%	93%	90%	94%
Learning Time (hours)	2	1.5	2.5	3	2.1	2.3	1.8
Reduction in Task Completion Time (min)	2	1.5	2.5	3	2.2	2.4	1.8
Output Clarity							
Output Accuracy (%)	92%	91%	89%	87%	90%	88%	93%
User Interpretation Score (%)	94%	92%	90%	86%	92%	90%	95%
Visual Quality Rating (1-5)	4.2	4.1	4	3.8	4.1	3.9	4.3
Task Efficiency							
Step Count	8	7	9	10	8	9	7
Processing Speed (min)	10 min	8 min	12 min	15 min	11 min	13 min	9 min
Completion Time Deviation (min)	±2 min	±1.5 min	±2.5 min	±3 min	±2 min	±2.5 min	±1.8 min
Customization and Flexibility							
Customization Options Count	6	5	4	3	5	4	6
User Satisfaction (1-5 Rating)	4.1	4	3.8	3.6	4.2	3.9	4.2
Ease of Customization (Time)	1.5 hours	1.3 hours	2 hours	2.5 hours	1.7 hours	2.2 hours	1.4 hours
Technical Support and Documentation							
Support Response Time (hrs)	24 hrs	20 hrs	26 hrs	30 hrs	22 hrs	25 hrs	21 hrs
Documentation Quality Rating (1-5)	4	4.2	4	3.7	4.1	3.8	4.3
User Confidence (%)	90%	89%	86%	84%	91%	87%	92%

v. Comparison of AI Tools

RealityCapture emerged as the top performer, excelling with a high user satisfaction rating of 4.4/5, swift navigation at 2.5 minutes, and efficient task completion in 13 minutes. Its notable accuracy of 91% and user-friendly interface make it a leading choice for effective and efficient architectural preservation. **AutoDesk ReCap** also demonstrated strong performance, boasting a user satisfaction rating of 4.3/5 and a rapid processing speed of 9 minutes. Its professional interface and excellent support make it a reliable tool for detailed preservation work. **Agisoft Metashape** provided high accuracy (92%) and good user satisfaction (4.2/5). However, it had a longer task completion time of 15 minutes. While it is effective, it requires more time compared to RealityCapture. **Pix4Dmapper** and **Colmap** showed balanced performance but were less efficient in processing speed and task completion. Pix4Dmapper had a user satisfaction rating of 4.1/5 and took 14 minutes to complete tasks, while Colmap had a rating of 4/5 and took 16 minutes. Both tools are practical but not as fast as RealityCapture or AutoDesk ReCap. **CycleGAN** had a steeper learning curve and lower user satisfaction at 3.8/5. Despite producing high-quality results, its complexity and slower processing time (20 minutes) limit its practicality for fast-paced projects. **3DF Zephyr** showed competent performance but had a longer task completion time (17 minutes) and a lower user satisfaction rating of 4/5. Its performance is solid but not as fast or user-friendly as the top-performing tools. In summary, RealityCapture and AutoDesk ReCap lead in user satisfaction and efficiency, making them ideal for fast and reliable architectural preservation. Agisoft Metashape, while accurate, requires more time. Pix4Dmapper, Colmap, and 3DF Zephyr offer balanced but less optimal performance, while CycleGAN, despite high-quality output, is less practical due to its complexity and processing time.

vi. Limitations and Benefits of AI Tools

RealityCapture stands out for its high user satisfaction (4.4/5), fast processing speed (2.5 min), and efficient task completion (13 min), though it has limited customization options. AutoDesk ReCap excels in technical support and processing speed (9 min) but has a complex interface and higher learning curve. Agisoft Metashape offers excellent accuracy (92%) and user satisfaction (4.2/5) but suffers from longer processing times (15 min). Pix4Dmapper provides a good balance of accuracy (90%) and user satisfaction (4.1/5), though it's not as fast as the top performers.

Colmap has decent user satisfaction (4/5) and reasonable processing speed (16 min) but is slower in task completion. 3DF Zephyr delivers solid performance with a user satisfaction rating of 4/5 but is less efficient and has a longer task completion time (17 min). CycleGAN excels in high-quality image restoration but has a steep learning curve and slow processing speed (20 min), making it less practical for quick projects. Each tool's mix of strengths and limitations affects its suitability based on specific project requirements and user needs.

vii. Implications

The findings suggest that for effective and efficient architectural preservation, tools like RealityCapture and AutoDesk ReCap are highly recommended due to their fast processing, high user satisfaction, and clear output. These tools enhance productivity and accuracy in the reconstruction and preservation of ancient architecture. CycleGAN and Agisoft Metashape, while powerful, present challenges in usability and speed. These tools are better suited for detailed and specific tasks where high accuracy is crucial, and time is less of a constraint. The results highlight the need for further development in AI tools to improve user-friendliness, reduce learning curves, and enhance processing speeds. Continued innovation in these areas will be essential for advancing the field of architectural preservation and maximizing the potential of AI technologies.

Tool / Criterion	UI Design	Navigation Efficiency	User Satisfaction	Error Rate	Ease of Use	Task Completion Rate	Time to Complete Tasks	Learning Curve	Output Clarity	Output Accuracy	Visual Quality	Customization Options	Support and Documentation		
Agisoft Metashape	4	3	4	3	4	4	3	3	4	4	4	3	4		
RealityCapture	4	4	4	3	4	4	4	3	4	4	4	3	4		
3DF Zephyr	3	3	3	3	3	3	3	2	3	3	3	3	3		
CycleGAN	3	3	3	2	3	3	3	2	3	3	3	2	3		
Pix4Dmapper	4	3	4	3	3	3	3	3	4	4	4	4	4		
Colmap	4	4	3	3	4	4	4	3	3	3	3	3	4		
AutoDesk ReCap	4	4	4	4	4	4	4	4	4	4	4	3	4		

1: Poor
2: Fair
3: Good
4: Very Good
5: Excellent

Fig 7. Heat Map

5. CONCLUSION

The evaluation of the AI tools for architectural preservation highlights several key findings. Agisoft Metashape and RealityCapture are highly recommended for their user satisfaction and efficient task completion. CycleGAN excels in image restoration but presents a steeper learning curve and lower user satisfaction. Pix4Dmapper and Colmap offer balanced performance but are less efficient in processing speed. AutoDesk ReCap provides excellent support but falls short in customization flexibility. Overall, selecting the right tool depends on the specific needs of the project and user expertise, with a combination of tools potentially offering the best results for comprehensive architectural preservation.

FUTURE ENHANCEMENTS

- ✓ **AI and AR/VR Integration:** Combining AI with augmented and virtual reality for immersive experiences.
- ✓ **Improved User Interfaces:** Developing more intuitive and user-friendly interfaces.
- ✓ **Enhanced Customization:** Providing more options for output customization.
- ✓ **Automated Workflow Optimization:** Using machine learning to streamline and optimize workflows.
- ✓ **Expanded Datasets:** Integrating larger and more diverse datasets to enhance model accuracy.

6. REFERENCES

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APPENDIX

Survey questions

User Interface Design

Navigation Efficiency:

1. How would you rate the ease of navigating through the tool's interface? (Rate from 1 to 5)
2. How much time did it typically take you to find the necessary features or tools?

User Satisfaction:

1. Overall, how satisfied are you with the tool's user interface? (Rate from 1 to 5)
2. What aspects of the interface did you find most or least satisfying?

Ease of Use

Task Completion Rate:

1. How consistently were you able to complete tasks using this tool? (Rate from 1 to 5)
2. Did you encounter any significant issues that prevented task completion?

Learning Curve:

1. How steep was the learning curve for this tool? (Rate from 1 to 5)
2. How much time did you need to become comfortable using the tool?

Initial Task Success Rate:

1. How successful were you in completing your first task using this tool? (Rate from 1 to 5)
2. Were there any initial challenges that affected your success rate?

Output Quality and Interpretation

Output Clarity:

1. How clear and understandable were the outputs generated by the tool? (Rate from 1 to 5)

2. Did the tool's outputs meet your expectations in terms of clarity?

Output Quality and Interpretation

Output Clarity:

3. How clear and understandable were the outputs generated by the tool? (Rate from 1 to 5)
4. Did the tool's outputs meet your expectations in terms of clarity?

Output Accuracy:

1. How accurate were the outputs produced by the tool compared to your expectations? (Rate on a scale of 1 to 5)
2. Were there any discrepancies or issues with the accuracy of the outputs?

User Interpretation:

1. How easy was it for you to interpret the results generated by the tool? (Rate on a scale of 1 to 5)
2. Did the tool provide adequate information to help you understand the results?

Task Efficiency

Processing Speed:

1. How would you rate the tool's processing speed when completing tasks? (Rate from 1 to 5)
2. Were there any delays or speed issues that affected your work?

Completion Time Deviation:

1. How consistent was the tool's task completion time compared to your expectations? (Rate from 1 to 5)
2. Did you notice any significant deviations from expected completion times?

Customization and Flexibility

Ease of Customization:

1. How easy was it to customize the tool to fit your specific needs? (Rate from 1 to 5)
2. Did the tool provide enough flexibility in customization options?

User Confidence:

1. How confident were you in using the tool after customizing it to your preferences? (Rate from 1 to 5)
2. Did the customization enhance your overall experience with the tool?

Technical Support and Documentation**Documentation Quality:**

1. How helpful was the tool's documentation in guiding you through its features? (Rate from 1 to 5)
2. Were there any gaps in the documentation that made it difficult to use the tool?

Support Response Time:

1. How responsive was the technical support when you encountered issues? (Rate from 1 to 5)
2. Did the support team resolve your issues in a timely manner?

INTERVIEW QUESTIONS

- Can you describe any specific challenges you encountered while using the tools?
- How did the tools compare to other digital tools you have used in the past?
- In your opinion, what improvements could be made to enhance the usability of these tools?
- How confident were you in interpreting the outputs generated by the tools?
- Do you think these tools could be effectively integrated into your workflow for architectural preservation? Why or why not?

TASK DESIGN

The usability tests consisted of three key tasks, designed to mimic real-world scenarios in architectural preservation:

Task 1: Generating a 3D model of a temple from a series of photographs using photogrammetry tools.

Task 2: Enhancing and restoring an old architectural photograph using CycleGAN.

Task 3: Creating a detailed 3D model from historical blueprints using AutoDesk ReCap

SAMPLE OUTPUT

